

5000+

CITIZENS REGISTERED

3

SYSTEM PHASES

7

CENSUS FORM STEPS

37

DATA FIELDS COLLECTED

12+

API ENDPOINTS

Abstract

JanSankhya is a multi-platform digital census management system designed to replace paper-based population data collection with a streamlined, technology-driven workflow. The system consists of a React-based administrative web portal, a Java Android mobile application for field enumerators, and a PHP-MySQL RESTful backend. Citizens are identified via unique QR-coded identity cards, enabling enumerators to instantly retrieve citizen profiles before recording a structured 37-field census covering education, employment, housing, family composition, health, migration, and government welfare enrollment. Three core innovations distinguish the system: duplicate detection; GPS location tagging; offline-first data architecture for low-connectivity rural environments with auto-sync.

Development Phases

PHASE 1Web Application

React + Vite + Tailwind CSS admin dashboard with role-based access, citizen registration, UID card printing, enumerator management, census entry portal, and live data records.

PHASE 2Android Application

Java Android app for enumerators with QR scanner, 7-step census form, offline storage using Room, GPS tagging, duplicate detection, and auto-sync on reconnection.

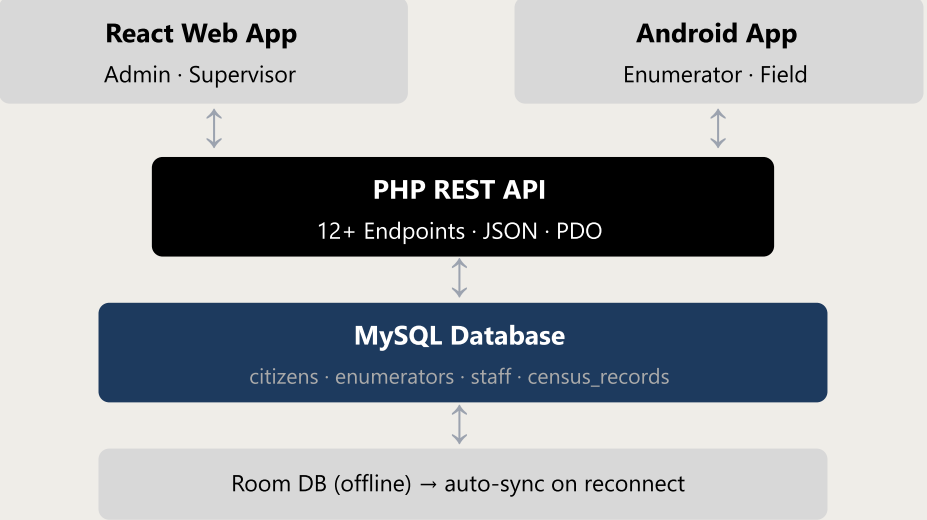
PHASE 3Backend API

PHP 8 RESTful API with PDO and MySQL, handling authentication, citizen management, census submission, enumerator statistics, and duplicate validation.

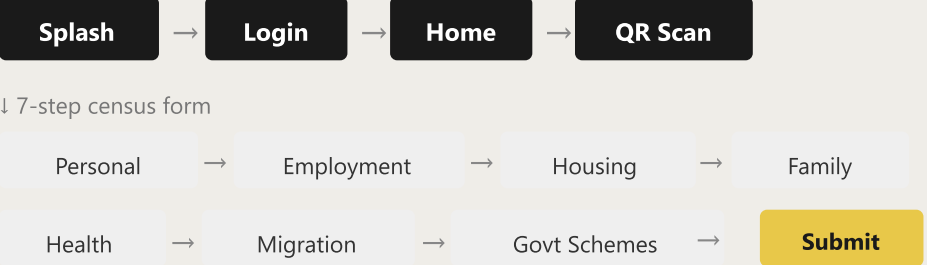
Technology Stack

Frontend	React 18, Vite, Tailwind
Mobile	Android Java, API 24+
Backend	PHP 8, PDO, XAMPP
Database	MySQL 8, Room (local)
HTTP	Volley, Fetch API
QR	ZXing, QRCode.js
Location	Google Play Services
Routing	React Router v6

System Architecture



Android Screen Flow



Database Schema



Key Innovations

Duplicate Detection

Before recording, the system cross-checks the citizen's UID against existing census_records. If a record already exists, the enumerator is notified immediately and the form is blocked — preventing double-counting without manual supervisor review.

GPS Location Tagging

Every submitted census record captures the enumerator's GPS latitude and longitude coordinates via Google Play Services FusedLocationProvider. This provides geographic proof of survey completion and enables spatial data analysis.

Offline-First Architecture

When internet connectivity is unavailable, census data is persisted locally using Android Room database. Upon reconnection, records are automatically detected and synchronized to the MySQL server in the background, ensuring zero data loss.

Web Portal Features

- Role-based access control for Admin and Supervisor with distinct navigation and permissions
- Citizen registration with auto-generated unique UID and scannable QR code
- Printable UID cards with real QR codes, batch export across multiple pages
- Enumerator and supervisor management with district and state assignment
- Census Records dashboard with search, filter, and detailed modal view
- Real-time statistics fetched from live database on login screen
- Data Generator for bulk citizen seeding with configurable parameters

Methodology

01 Requirement Analysis

Identified inefficiencies in manual census workflows; defined functional requirements for Admin, Supervisor, and Enumerator roles.

02 Schema Design

Designed normalized relational database with referential integrity between citizens, enumerators, staff, and census records.

03 API Development

Built stateless RESTful PHP endpoints with PDO, CORS headers, and structured JSON responses for cross-platform compatibility.

04 Parallel UI Development

Web and Android apps developed concurrently with iterative testing, consistent color system, and shared API contracts.

05 End-to-End Testing

QR scanning, GPS capture, offline sync, duplicate detection, and form submission tested across multiple real Android devices.

Conclusion & Future Scope

What was achieved

JanSankhya successfully demonstrates a complete, working alternative to paper-based census collection. The system operates across three platforms with a shared data layer, supports offline data collection, prevents duplicate records, and provides administrators with live visibility into census progress — capabilities unavailable in traditional field surveys.

Technical Outcomes

- Fully functional end-to-end data pipeline from citizen scan to database record
- Offline Room database with background synchronization on reconnect
- Real QR code generation using ZXing and QRCode.js libraries
- GPS coordinates stored per record for geographic audit trail

Future Enhancements

- Cloud deployment for network-independent access from any location
- Multilingual support for regional language data entry
- Power BI or embedded analytics for real-time population insights
- Biometric verification layer for enumerator authentication in the field

PROJECT SUMMARY

System	Digital Census Management
Platforms	Web · Android · REST API
Stack	React · Java · PHP · MySQL
Citizens	5000+ registered
API Endpoints	12+
Form Fields	37 data fields
University	Nirma University